DEUTSCHE NATIONAL BIBLIOTHEK

Workshop

Evaluating Automated Subject Indexing Methods

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Welcome

Welcome to the Workshop on **Evaluating Automated Subject Indexing Methods**

DEUTSCHE NATIONAL BIBLIOTHEK

About me:

- Maximilian Kähler, German National Library (DNB)
- education: mathematics & theoretical physics
- project lead for DNB-AI-Project on advancing automated subject indexing

About this Workshop:

- Explore evaluation techniques for evaluating automated subject indexing methods, focussing on practical applications using the **CASIMiR**[1] package.
- Created as part of DNB-AI-Project, funded by the German Federal Commissioner for Culture and Media (BKM) [2]
- All workshop materials are available online[3]

1. https://github.com/deutsche-nationalbibliothek/casimir
2. https://kulturstaatsminister.de/#
3. https://github.com/deutsche-nationalbibliothek/casimir-workshop

Introduction

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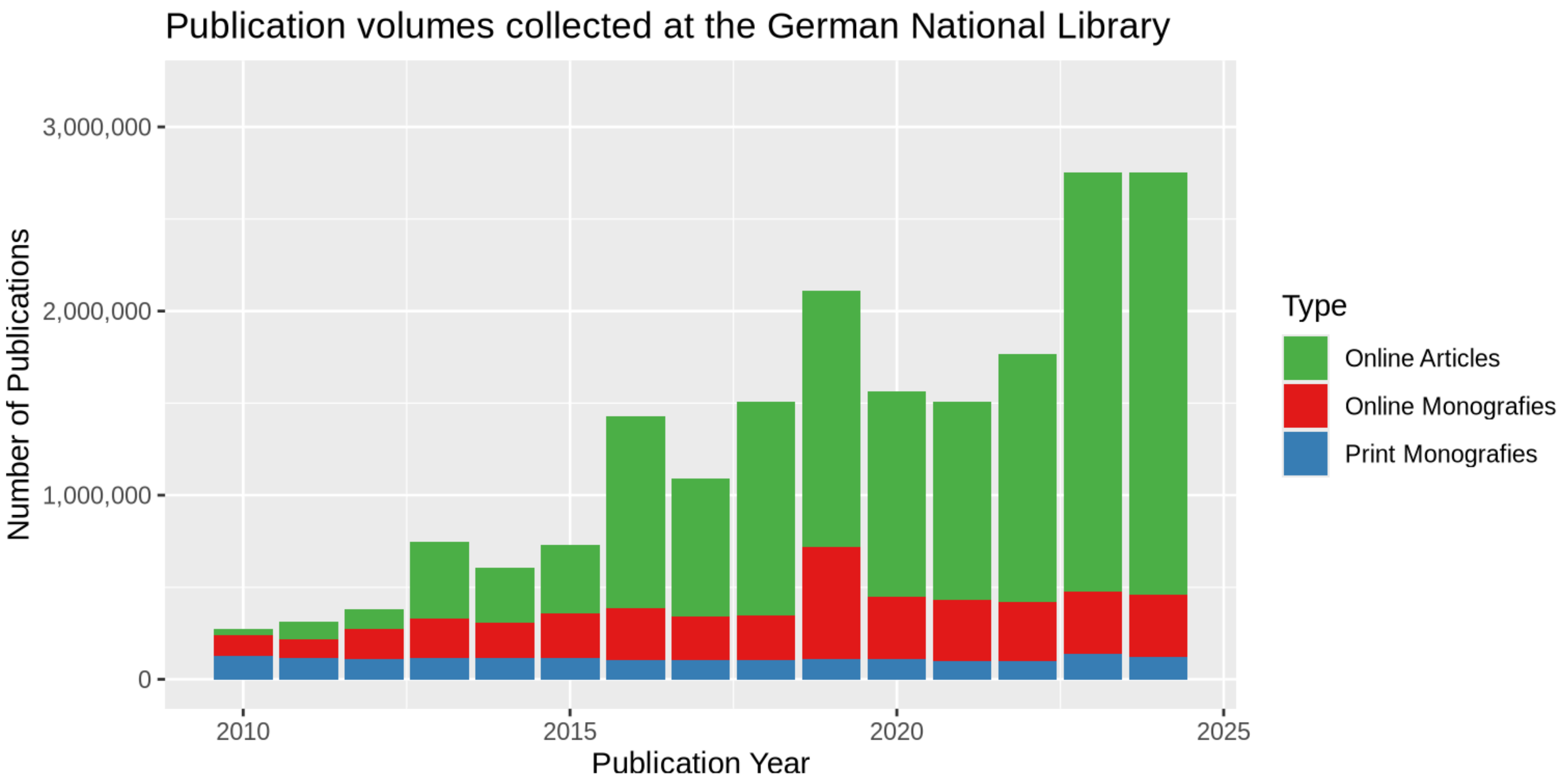
Why Automated Subject Indexing?



- increasing publication volume
- limited human resources for manual subject indexing
- need for consistent and scalable indexing methods

Automated Subject Indexing@DNB

- In production since 2014
- Complete refactorization of software architecture in 04/2022 [1]
- Core component is the open source library annif [2]
- Annual throughput of ~170.000 publications per year
- Our target vocabulary is the Integrated Authority File[3] (GND) containing >1.4M potential concepts

1. Poley etal. 2025. “Automatic Subject Cataloguing at the German National Library.” https://doi.org/10.53377/lq.19422
2. Suominen 2019. “Annif: DIY Automated Subject Indexing Using Multiple Algorithms.” https://doi.org/10.18352/lq.10285.
3. https://gnd.network/

Why subject Indexing is hard

- very large Vocabularies like GND or LCSH (thousands to millions of labels)
- very sparse annotations (only few correct labels per document)
- highly imbalanced label distributions (some labels are very frequent, others very rare)
- incomplete ground truth (not all correct labels are assigned in manual indexing)
- hierarchical relationships between labels (some labels are more general/specific than others)
- …

Improving Automated Subject Indexing

Three complementary approaches:

Find and write better algorithms for automated subject indexing

Reduce the complexity of the problem

Get better at diagnosing good keyword extraction

Methods for Subject Indexing

An Overview

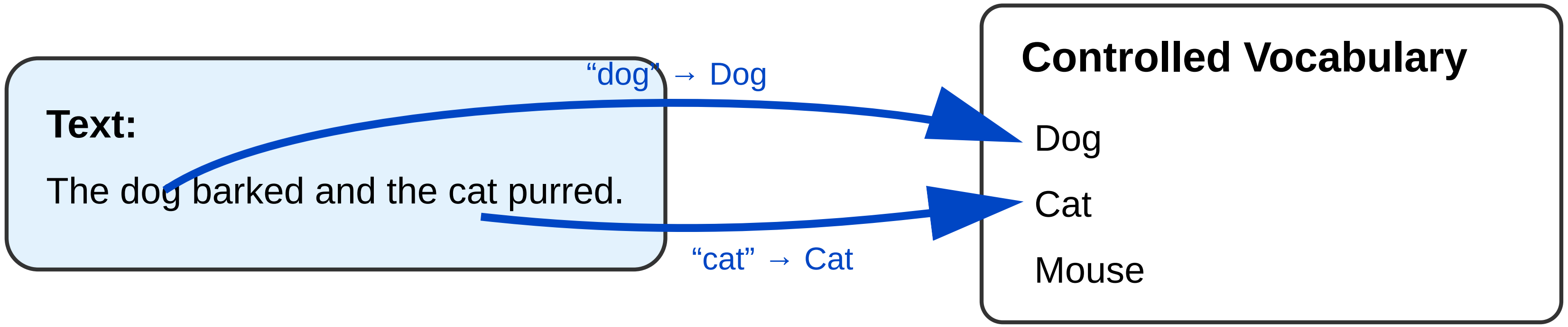
Before the advent of Deep Learning and transformers:

- Lexical Matching [1,2]
- Statistical Methods for Extreme Multi-Label Classification (XMLC):[3,4]
  - 1vsAll classifiers [5]
  - Partitioned Label Trees [6,7,8]

2 and 6 are backends of annif

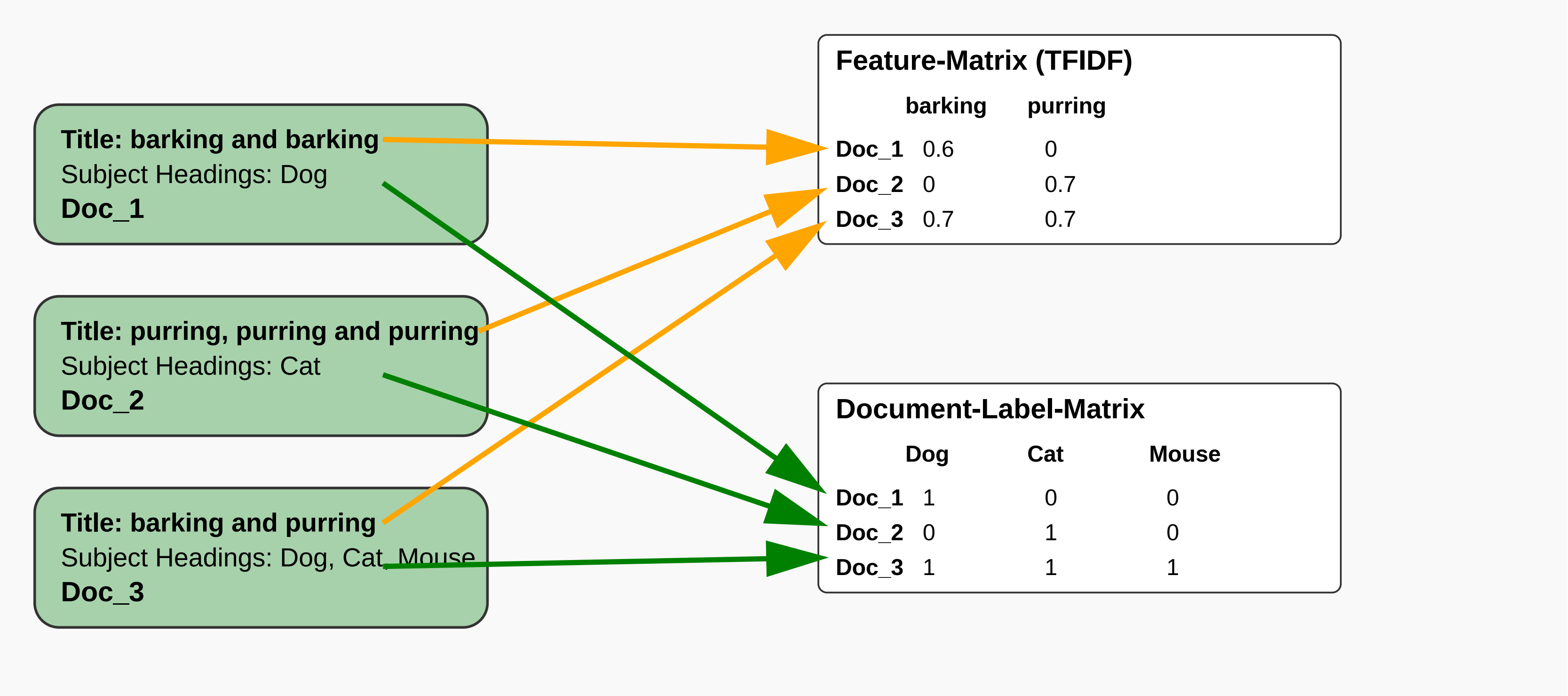
1. Medelyan, O., Frank, E., & Witten, I. H. (2009). Human-competitive tagging using automatic keyphrase extraction.
2. Maui-like Lexical Matching https://github.com/NatLibFi/Annif/wiki/Backend%3A-MLLM
3. Bhatia etal. 2016. “The Extreme Classification Repository: Multi-Label Datasets and Code.” http://manikvarma.org/downloads/XC/XMLRepository.html.
4. Dasguptaetal. 2023. “Review of Extreme Multilabel Classification,” February. https://arxiv.org/abs/2302.05971v2.
5. Schultheis, E. and Babbar, R. (2021) “Speeding-up One-vs-All Training for Extreme Classification via Smart Initialization.” Available at: https://doi.org/10.48550/arxiv.2109.13122
6. Khandagale, S., Xiao, H., & Babbar, R. (2020). Bonsai: diverse and shallow trees for extreme multi-label classification.
7. Prabhu, Y., Kag, A., Harsola, S., Agrawal, R., & Varma, M. (2018). Parabel: Partitioned label trees for extreme classification with application to dynamic search advertising
8. Omikuji Rust-Library: https://github.com/tomtung/omikuji

Lexical Matching



- can match any keyword in the vocabulary
- heuritstics like frequency, position, etc. are used to rank the keywords after matching

Statistical Methods



- statistical apporaches learn correlations between **feature matrix** and **document-label matrix**
- statistical apporaches don't need prefLabels: they do not interpret the description of labels

After the invention of transformers I

Refitting old methods with transformers:

- Lexical Matching → Embedding Based Matching [1]
  - performs matching in embedding space rather than lexically
- Statistical Methods:
  - Partitioned Label Trees → X-Transformer [2]
  - represent document features with transformer embeddings rather than only TFIDF

1. DND-Prototype: https://github.com/deutsche-nationalbibliothek/ebm4subjects
2. Zhang etal. (2021). Fast Multi-Resolution Transformer Fine-tuning for Extreme Multi-label Text Classification. https://arxiv.org/abs/2110.00685v2

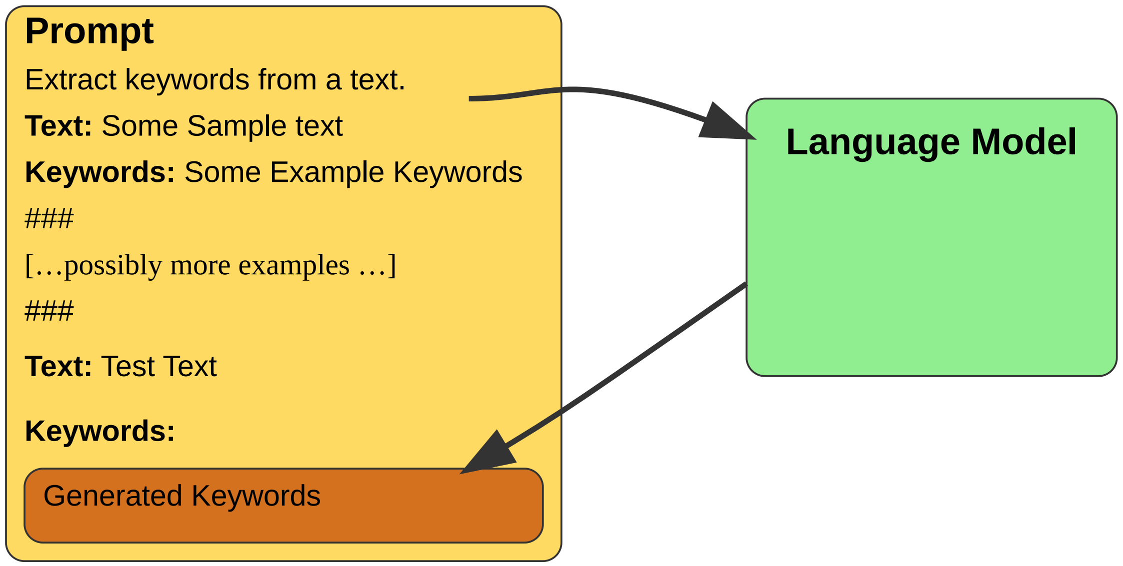
After the invention of transformers II

Going for generative LLM-powered methods:

- prompt LLMs to generate keywords for documents
- use few-shot prompting to guide LLMs towards desired behavious
- map generated keywords to controlled vocabulary

`Few-Shot-Prompting` + `Mapping`

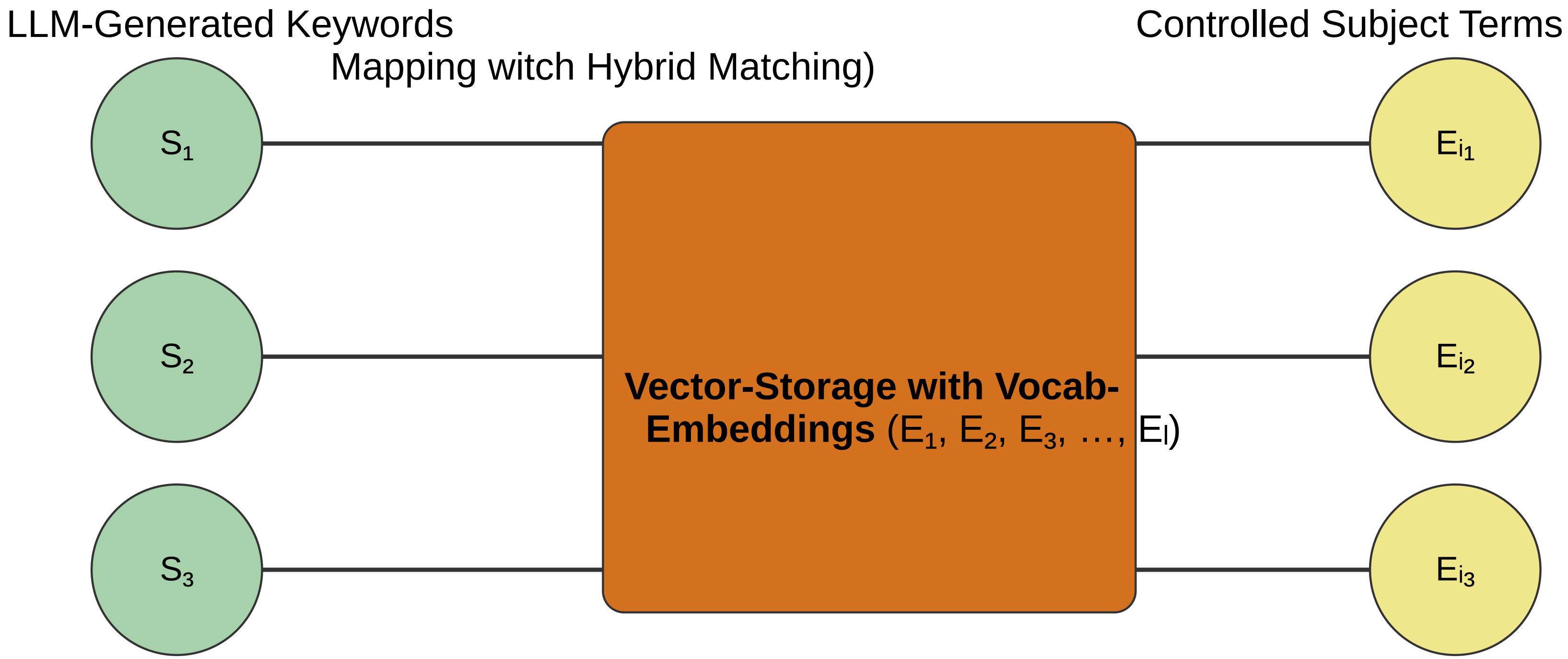
The idea of few shot prompting



- exploits world knowledge in LLMs
- needs only few training examples

Mapping

- LLM has no knowledge of controlled vocabulary
- extra step maps generated keywords to controlled vocabulary
- hybrid search: lexical matching + embedding based matching powered by vector storage



Advanced LLM-based methods

Few-shot prompting and mapping alone do not yield satisfactory results yet

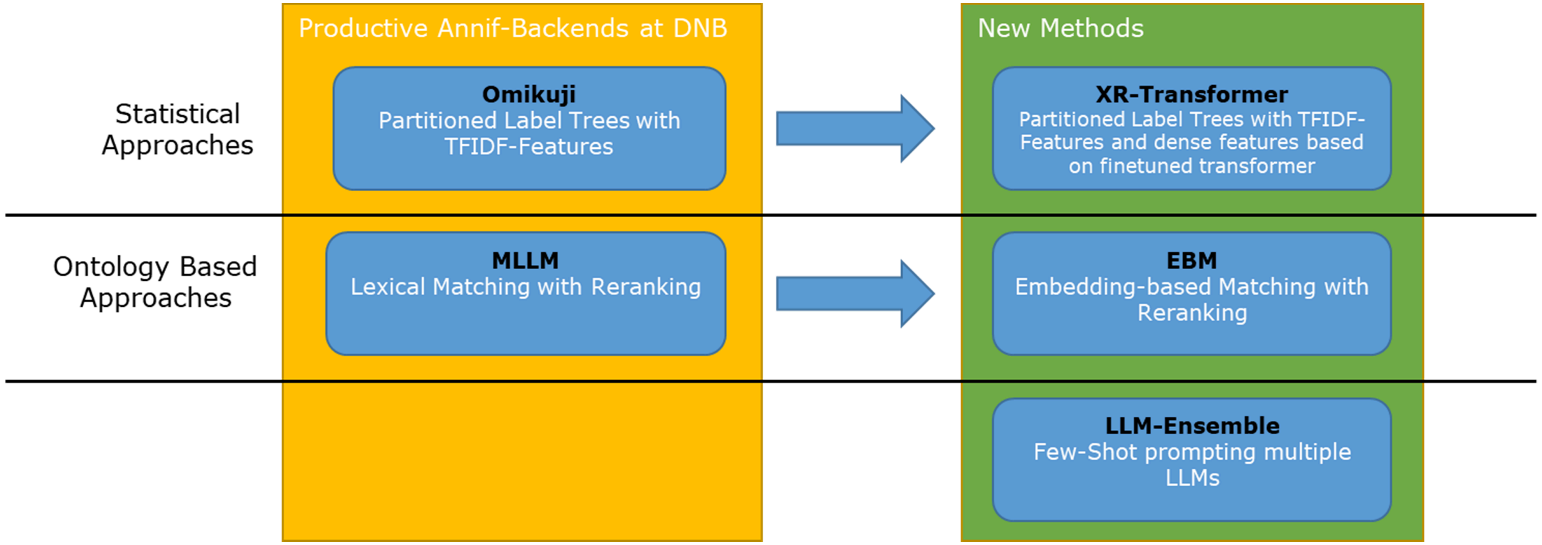
–> Advanced methods:

- combine multiple LLM-based methods into an **LLM-ensemble**[1]
- use LLMs to Rank and filter candidate labels from multiple methods
- use retrieval to augment LLM prompts with relevant context[2]

1. Kluge, L. and Kähler, M. (2025) “DNB-AI-Project at SemEval-2025 Task 5: An LLM-Ensemble Approach for Automated Subject Indexing,” https://aclanthology.org/2025.semeval-1.148/.
2. Kähler, M., Kluge, L. and Konermann, K. (2025) “DNB-AI-Project at the GermEval-2025 LLMs4Subjects Task: KIFSPrompt - Knowledge-Injected Few-Shot Prompting,” https://aclanthology.org/2025.konvens-2.42/.

Methods: Summary

5 Methods we work with today:



Your task: identify which method is behind which anonymised name[1]

Artful Accordion | Bold Bassoon | Charming Cello | Dreamy Didgeridoo | Embracing Euphonium

1. Icons created by Freepik - Flaticon

Datasets for this Workshop

Titles and Vocabulary

Document Titles:

- German National Library (DNB) Test Set
  - 8.415 document titles with manual subject indexing
  - German scientific literature from 18 selected subject groups in equal proportions
- Available at: `data/` folder in this repository
- `data/README.md` contains descriptions of all dataset files

Vocabulary:

- ~200K unique concepts, subsetted from the Integrated Authority File (GND)[1]
- each subject heading has a unique identifier (`label_id`) and a preferred label (`label_text`)
- the GND includes subject headings, geographic headings, personal names, corporate names, and more

1. https://gnd.network/

Predictions

- methods are anonymized for unbiased evaluation
- Predictions from 5 different methods provided under `data/test-set_predictions/`:
  - `artful-accordion.csv`
  - `bold-bassoon.csv`
  - `charming-cello.csv`
  - `dreamy-didgeridoo.csv`
  - `embracing-euphonium.csv`
- key-columns are `doc_id` for document identifiers and `label_id` for GND subject headings, which are expected by CASIMiR functions
- up to 100 suggested doc-label pairs per method and document
- every suggested label also has a `score` between 0 and 1 indicating the confidence of the method

Workshop Game:

Can you guess which method is behind which file name?

Practical I: Manual Inspection

Practical: Go to `workbooks/01_inspecting-results-manually.qmd` to inspect the data formats and datasets provided.

Warning: English translations of document titles and GND subject labels are created by generative AI and provided for convenience only. They may contain errors or inaccuracies. Subject-Suggestions are all based on subject indexing methods applied to German texts and German labels.

💡 If you speak German use the `_ger` column names in the code examples instead of the `_eng` variants to see the original German texts.

Practical I: Example

Qualitative Method Comparison							
label_id	label_text	gold	score_artful-accordion	score_bold-bassoon	score_charming-cello	score_dreamy-didgeridoo	score_embracing-euphonium
1166742806 - Inclusive School and Curriculum Development: From Aspiration to Successful Implementation							
950251194	Transformation	FALSE	0.05997121	NA	NA	NA	NA
041316657	Claim	FALSE	0.31719807	NA	NA	0.1859743	NA
04126892X	School development	TRUE	NA	NA	0.11813986	0.4799480	0.858
041351487	Organizing the Class	TRUE	NA	NA	NA	NA	NA
1000723437	Inclusive School	TRUE	NA	0.2927039	0.04837416	0.7240117	0.066
965002845	Inclusion (Sociology)	TRUE	NA	0.1999691	0.06500251	NA	0.148
041276612	School development planning	FALSE	NA	0.1276198	NA	NA	NA
04053474X	School	FALSE	NA	0.1385269	NA	NA	NA
100072185X	Inclusive Pedagogy	FALSE	NA	0.1656229	0.04911679	0.6998804	0.041
041351754	Educational research	FALSE	NA	NA	0.04595719	NA	NA
123322929X	Inclusive teaching	FALSE	NA	NA	NA	0.3014980	NA
040118827	Germany	FALSE	NA	NA	NA	NA	0.099

Evaluation of Subject Indexing Methods

Metric Basics

We work mostly in the **binary relevance scenario**: Comparing predictions to a goldstandard

	Goldstandard yes	Goldstandard no	
Prediction yes	true positives (`tp`)	false positives (`fp`)	$\mathrm{Prec} = \frac{tp}{tp+fp}$
Prediction no	false negatives (`fn`)	true negatives (`tn`)	
	$\mathrm{Rec} = \frac{tp}{tp+fn}$		

Recall

How many of the expected gold standard labels were actually suggested?

Precision

How many of the suggested labels are actually correct?

Example: Document Level Precision and Recall

Consider the following example document title:

> Media habitus and biographical legend -
> Writerly performance practices in the age of digitalization

label_id	label_text	gold-standard	suggested
041305450	Authorship	TRUE	TRUE
040359646	Literature	FALSE	TRUE
040272230	Performance	FALSE	TRUE
040533093	Writer	FALSE	TRUE
041230655	Digitalization	FALSE	TRUE
041223497	Self-presentation	TRUE	FALSE

- tp = 1 (Authorship)
- fp = 4 (Literature, Performance, Writer, Digitalization)
- fn = 1 (Self-presentation)
- tn = Vocab-Size - 6 (all other labels not suggested and not in goldstandard)

$$\mathrm{Prec} = \frac{1}{1+4} = 0.2$$

$$\mathrm{Rec} = \frac{1}{1+1} = 0.5$$

Metrics Basics: continued

- Precision and Recall can be computed on different aggregation levels:
  - per document
  - per label
  - overall (micro/macro averaged)
- Precision and Recall can be combined into F1-Score:

$$\mathrm{F1} = 2 \cdot \frac{\mathrm{Prec} \cdot \mathrm{Rec}}{\mathrm{Prec} + \mathrm{Rec}}$$

- In subject indexing, when results are ranked by a confidence score,we are often interested in top-k suggestions:
  - Precision@k, Recall@k, F1@k
  - e.g. how many of the top-5 suggestions are correct?

Practical II

Practical: Go to `workbooks/02_set-retrieval-metrics.qmd` to compute retrieval metrics for the provided subject indexing methods.

Example: Compute Precision, Recall, and F1-Score for `artful-accordion` method

```
compute_set_retrieval_scores(
  predicted = predictions[["artful-accordion"]],
  gold_standard = gold_standard,
  k = 5, # limit to top-5 suggestions
  rename_metrics = TRUE
)
```

```
# A tibble: 4 × 4
  metric  mode    value support
  <chr>   <chr>   <dbl>   <dbl>
1 f1@5    doc-avg 0.277    8415
2 prec@5  doc-avg 0.289    8197
3 rec@5   doc-avg 0.349    8415
4 rprec@5 doc-avg 0.414    8197
```

Practical III: Dimensions of Evaluation

Datasets often have multiple **document-based** dimensions that can be used for stratified evaluation:

- subject groups
- document types
- languages
- publication years

Alternatively: Stratify along **vocabulary-based** dimensions:

- entity types (e.g., topics, places, persons)
- subject groups
- hierarchical levels
- frequent vs. rare labels

Practical: Go to `workbooks/03_stratified-set-retrieval-metrics.qmd` to explore stratified evaluation using different dimensions of the provided datasets.

Practical III: Example

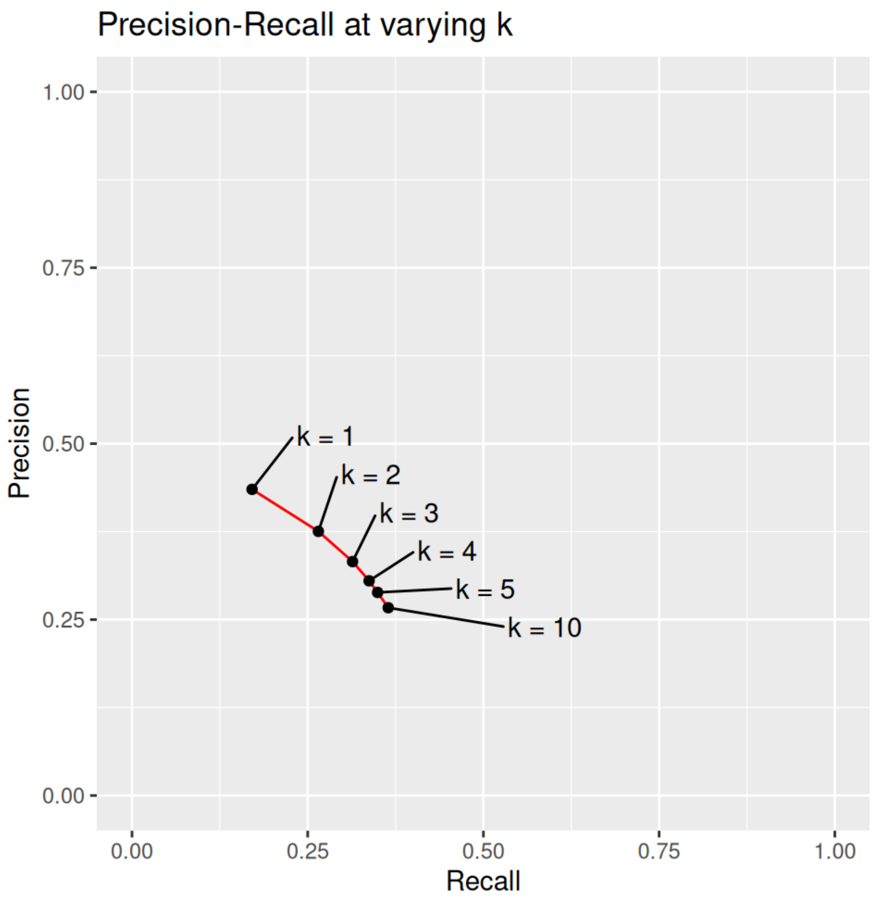
Given the mapping table `subject_groups` that assigns each document to a subject group, we can compute stratified retrieval scores:

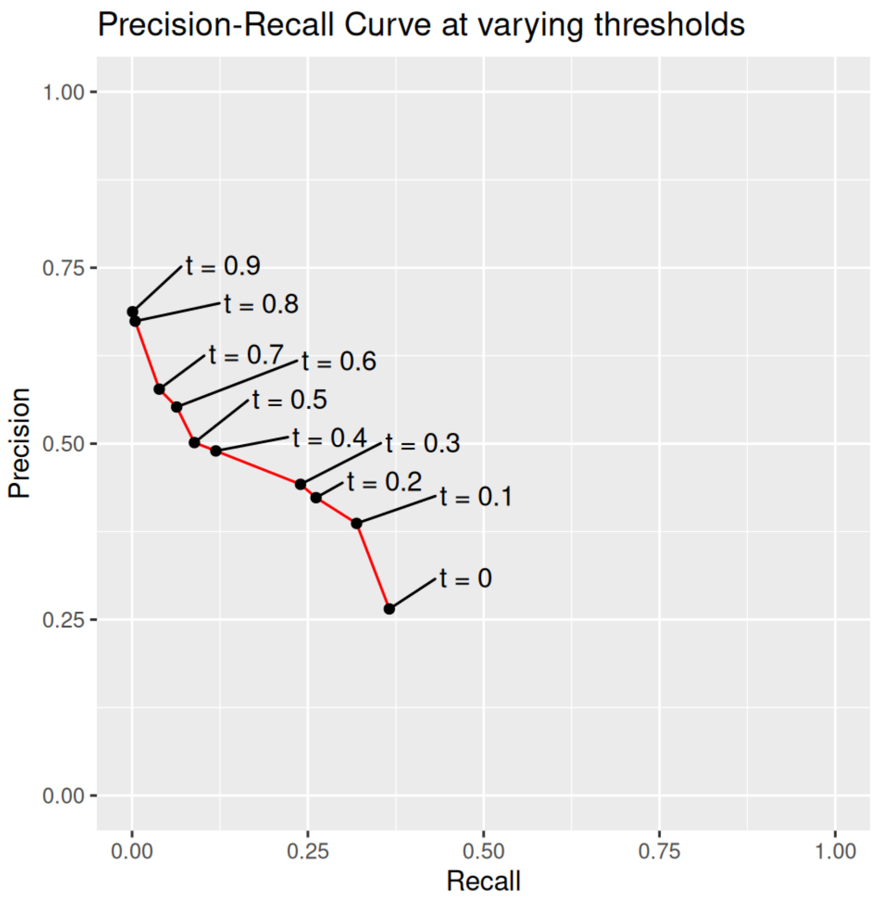
```r
res_at_5_by_sg_artful_accordion <- compute_set_retrieval_score
  predicted = predictions[["artful-accordion"]],
  gold_standard = gold_standard,
  doc_groups = subject_groups,
  k = 5
)

kable(head(res_at_5_by_sg_artful_accordion, 3))
```

sg	sg_label_ger	sg_label_eng	metric	mode	value	support
004	Informatik	Computer Science	f1	doc-avg	0.2111046	500
100	Philosophie	Philosophy	f1	doc-avg	0.3383969	500
150	Psychologie	Psychology	f1	doc-avg	0.2642708	500

Precision-Recall Curves

So far: set retrieval metrics at fixed `k = 5`





Precision-Recall curves show the trade-off between precision and recall at different thresholds

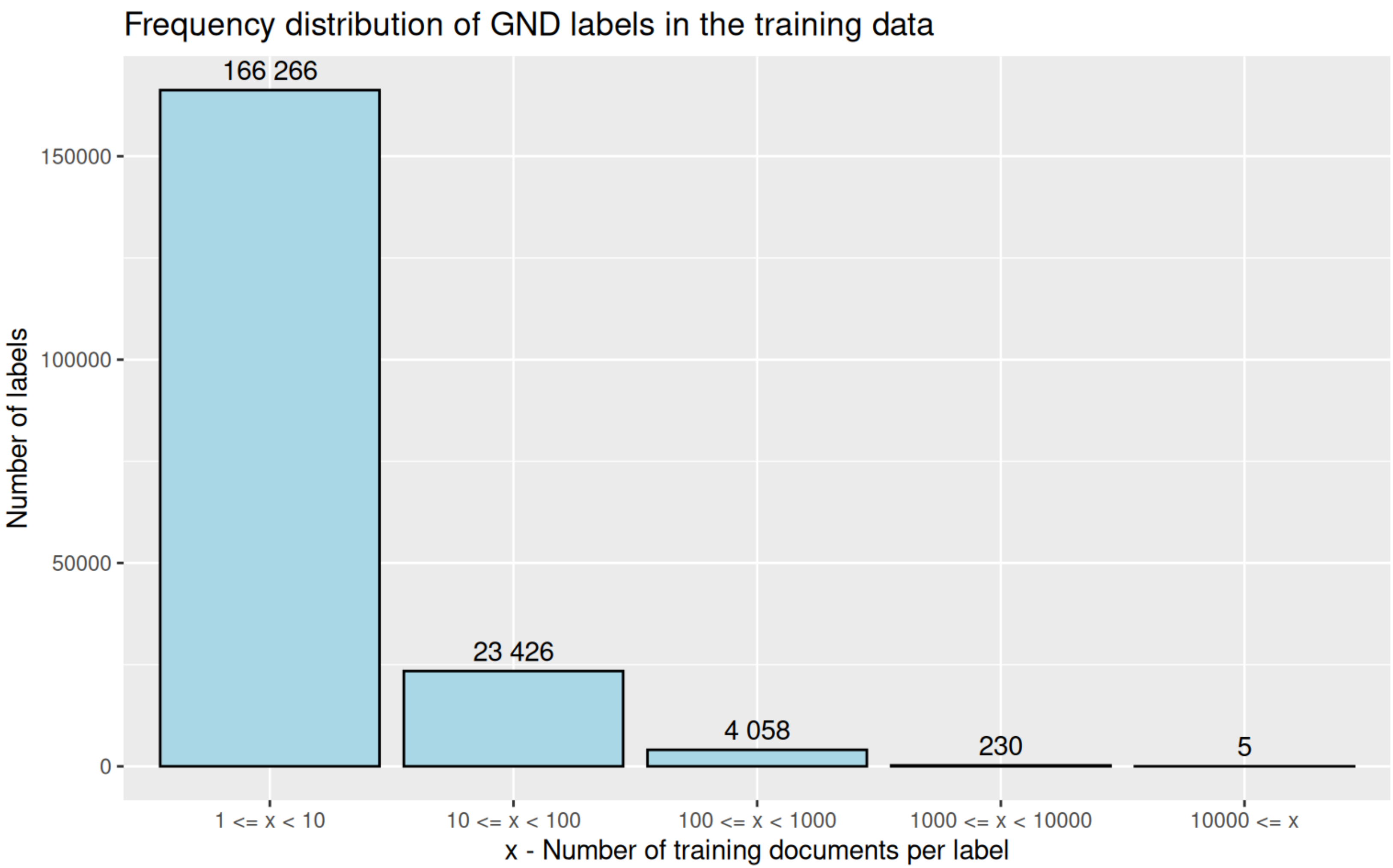
Practical IV: Precision Recall Curves

Practical: Go to `workbooks/04_precision-recall-curves.qmd` to compute precision-recall curves for the provided subject indexing methods.

Example: Compute Precision-Recall Curve with CASIMiR

```
pr_curve <- compute_pr_curve(
  predicted = predictions[["artful-accordion"]],
  gold_standard = gold_standard,
  steps = 10 # number of thresholds to evaluate
)

ggplot(pr_curve$plot_data, aes(x = rec, y = prec_cummax)) +
  geom_point() +
  geom_path() +
  coord_fixed(xlim = c(0, 1), ylim = c(0, 1)) +
  ggtitle("Precision-Recall Curve for artful-accordion, comput
```

Analysing the long tail



Training data: ~ 950K book titles from DNB with manual GND subject annotations

Practical V: Long Tail Analysis

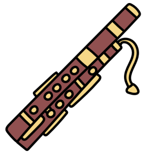
Practical: Go to `workbooks/05_analysing-the-long-tail.qmd` to analyse performance of subject indexing methods on frequent vs. rare labels.

Practical VI: Final Assignment

Practical: Go to `workbooks/06_final-assignment.qmd` and try to solve todays quiz:

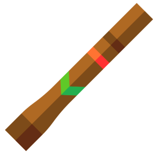
Your task: identify which method is behind which anonymised name[1]

Artful Accordion



Bold Bassoon

Charming Cello



Dreamy Didgeridoo

Embracing Euphonium

1. Icons created by Freepik - Flaticon

Solution: Methods Revealed

Instrument	Method Name
Artful Accordion	Lexical Matching
Bold Bassoon	LLM-Ensemble
Charming Cello	X-Transformer
Dreamy Didgeridoo	Embedding Based Matching
Embracing Euphonium	Omikuji (TFIDF-based Statistical Method)

Haven’t found your favourite method?

Consider combining them into **ensembles** for improved performance![1]

1. Orchestra icons created by mia elysia - Flaticon

Advanced Topics

Bonus: Conditional label weights

Do true positives, false positive and false negatives mean the same to us?

We can assign different **costs** to different types of errors:

	Goldstandard yes	**Goldstandard no**
Prediction yes	$C_{\mathrm{tp}} \cdot \mathrm{tp}$	$C_{\mathrm{fp}} \cdot \mathrm{fp}$
Prediction no	$C_{\mathrm{fn}} \cdot \mathrm{fn}$	$C_{\mathrm{tn}} \cdot \mathrm{tn}$

Assume **label dependent costs** for true positives and false negatives:

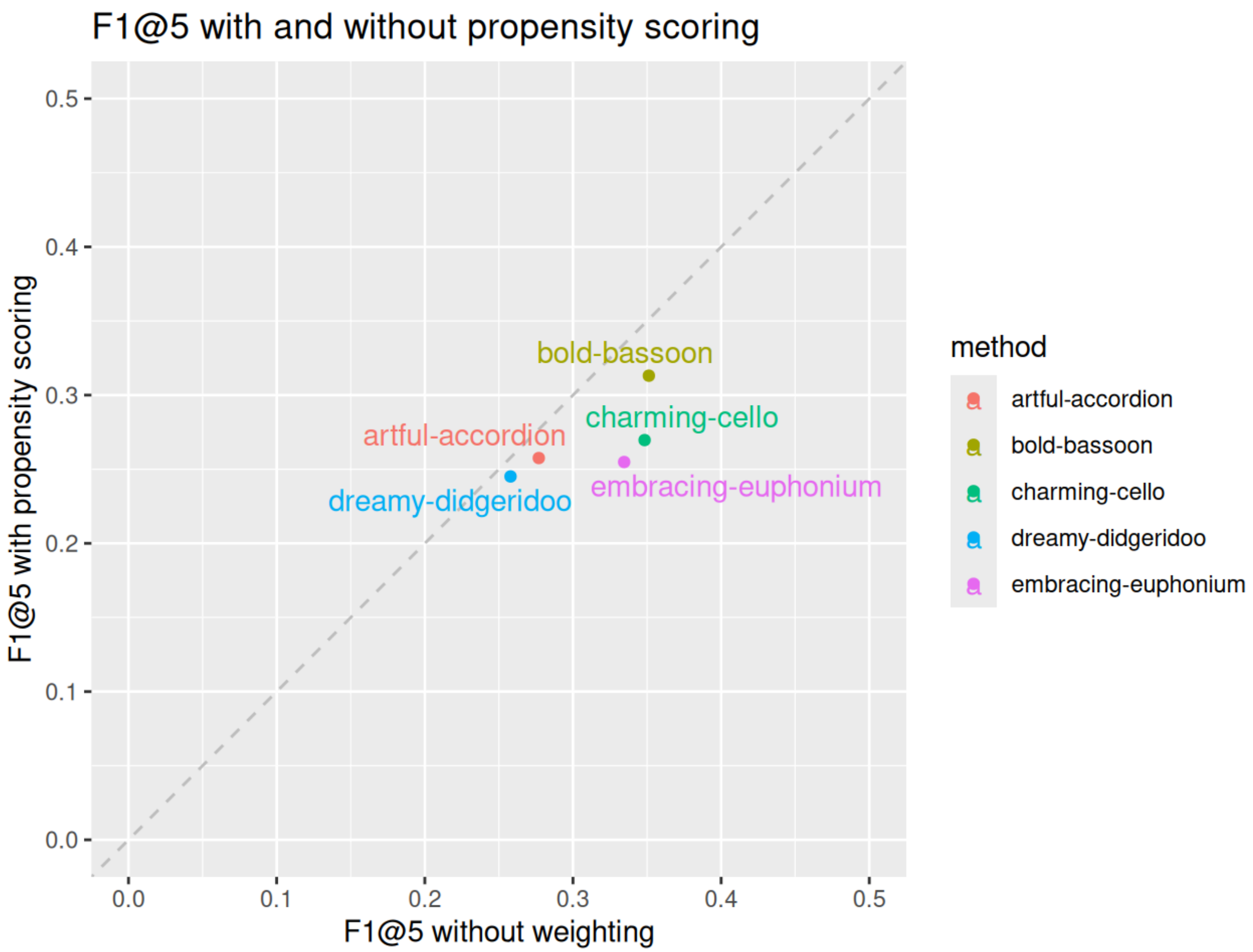
$$C_{\mathrm{tp}} = C_{\mathrm{fn}} = w_{\lambda},$$

and constant costs for false positives:

$$C_{\mathrm{fp}} = const$$

where w_{λ} is a weight depending on label λ, inversely proportional to its frequency in the training data.

Bonus: Conditional label weights II



Bonus: Beyond binary relevance I

Revisit the earlier example document title:

> Media habitus and biographical legend -
> Writerly performance practices in the age of digitalization

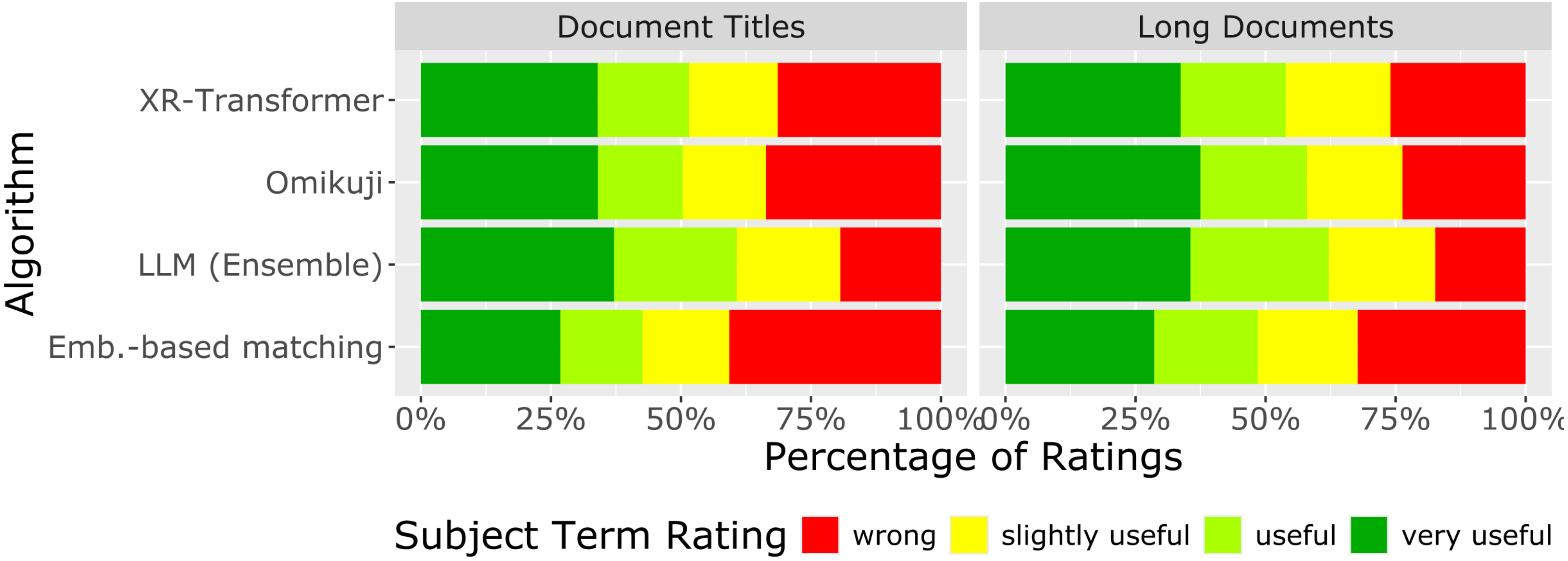
label_id	label_text	gold-standard	suggested
041305450	Authorship	TRUE	TRUE
040359646	Literature	FALSE	TRUE
040272230	Performance	FALSE	TRUE
040533093	Writer	FALSE	TRUE
041230655	Digitalization	FALSE	TRUE
041223497	Self-presentation	TRUE	FALSE

- Incomplete ground truth: not all correct labels are assigned in manual subject indexing
- false positives may actually still be helpful suggestions for retrieval

Bonus: Beyond binary relevance II

Graded relevance levels can be used to model this, e.g.:

Ordinal Relevance Level
very helpful
helpful
slightly helpful
wrong



Generalized Precision and Recall

Graded relevance leads to the notion of **generalised precision** and **generalised recall**[1]

Ordinal Relevance Level	Metric Relevance Level r
very helpful	1
helpful	2/3
slightly helpful	1/3
wrong	0

Generalised Precision

$$\frac{tp + \Delta_{rel}}{tp + fp}$$

Generalised Recall

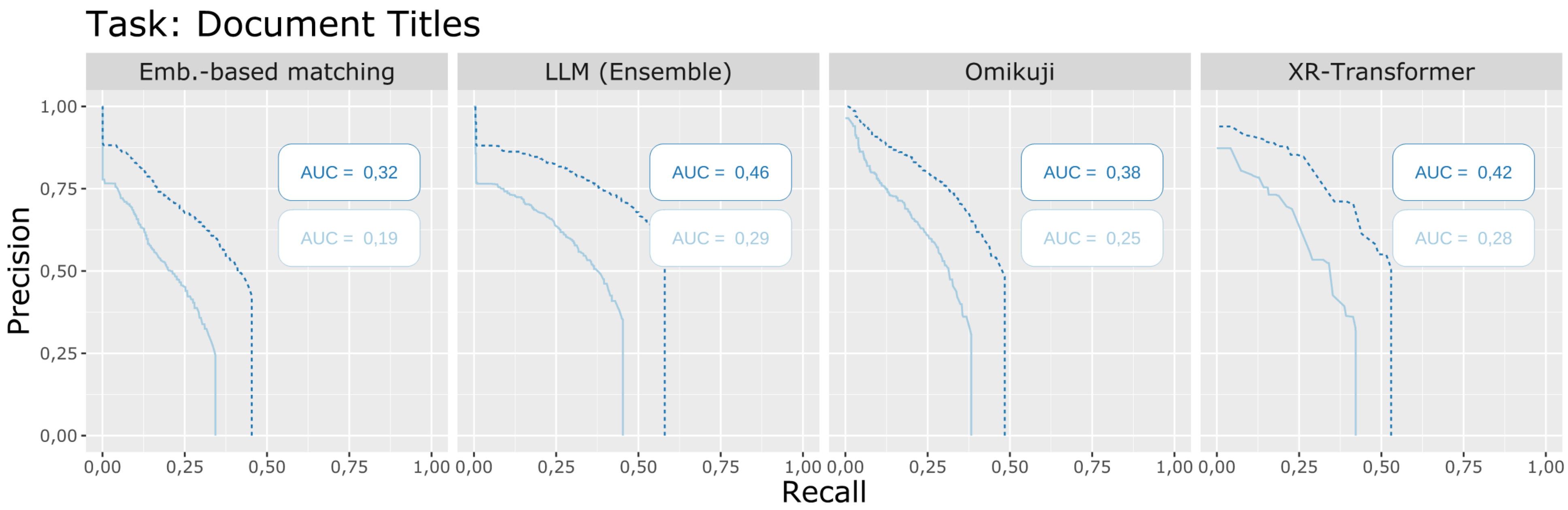
$$\frac{tp + \Delta_{rel}}{tp + fn + \Delta_{rel}}$$

with $\Delta_{rel} = \sum_{i \in false\ positives} r_i$ with metric relevance ratings $0 \leq r_i \leq 1$

1. Kekäläinen, Jaana, and Kalervo Järvelin. 2002. "Using Graded Relevance Assessments in IR Evaluation." Journal of the American Society for Information Science and Technology 53 (November): 1120–29. https://doi.org/10.1002/asi.10137.

Generalized Precision and Recall Curves

Generalised precision-recall curves can be computed similarly to standard precision-recall curves (also supported in CASIMiR):



Discussion

Discussion

Your opinion:

What other aspects of subject indexing methods should we evaluate?

The End

Thank you!

Contact

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